

Администрирование локальных сетей

Лабораторная работа №9

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Цели и задачи работы

Изучение протокола STP и его модификации Rapid PVST+, настройка резервирования и агрегирования каналов.

Выполнение лабораторной работы

Формирование резервного соединения

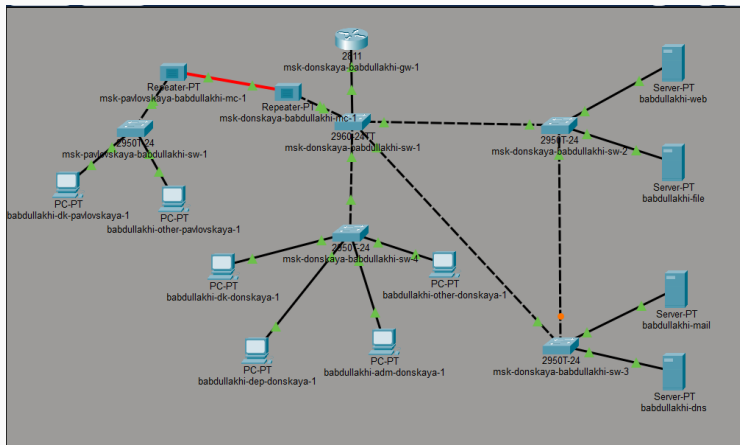


Рис. 1: Резервное соединение

Проверка прохождения ICMP

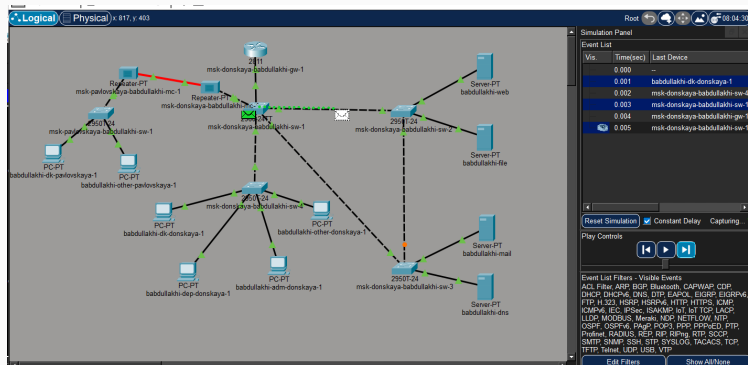


Рис. 2: ICMP трафик

```
msk-donskaya-babdullakhi-sw-2#sh spa
msk-donskaya-babdullakhi-sw-2#sh spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0001.C950.D2B0
             Cost        23
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID  Priority    32771 (priority 32768 sys-id-ext 3)
             Address     000D.BDB2.2603
             Hello Time  2 sec   Max Age 20 sec   Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/2                    Desg FWD 19        128.2    P2p
Fa0/1                    Desg FWD 19        128.1    P2p
Gi0/1                    Root FWD 4         128.25   P2p
Gi0/2                    Desg FWD 4         128.26   P2p

msk-donskaya-babdullakhi-sw-2#
```

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Рис. 3: STP состояние

Назначение root-коммутатора

```
msk-donskaya-babdullakhi-sw-1(config)#spa
msk-donskaya-babdullakhi-sw-1(config)#spanning-tree vlan 3 root prim
msk-donskaya-babdullakhi-sw-1(config)#exit
msk-donskaya-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-1#show s
msk-donskaya-babdullakhi-sw-1#show sp
msk-donskaya-babdullakhi-sw-1#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority      24579
             Address      000A.F3DB.58D3
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority      24579  (priority 24576 sys-id-ext 3)
             Address      000A.F3DB.58D3
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/1          Desg FWD 4        128.25   P2p
Gi0/2          Desg FWD 4        128.26   P2p
Fa0/24         Desg FWD 19       128.24   P2p
Fa0/23         Desg FWD 19       128.23   P2p
Fa0/1          Desg FWD 19       128.1    P2p

msk-donskaya-babdullakhi-sw-1#
```

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Изменение маршрутов трафика

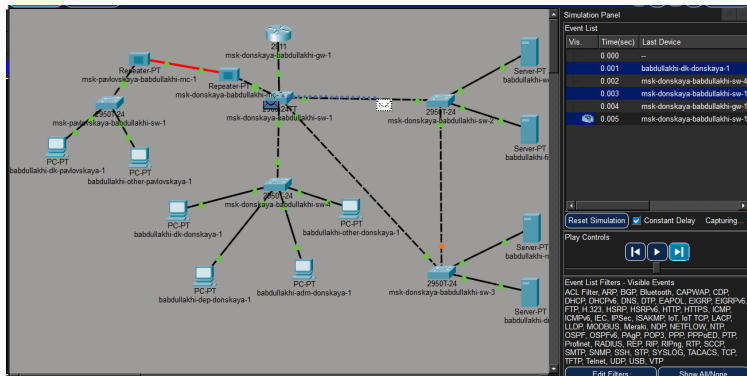


Рис. 5: Маршрутизация пакетов

```
msk-donskaya-babdullakhi-sw-2#
msk-donskaya-babdullakhi-sw-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-babdullakhi-sw-2(config)#int f0/1
msk-donskaya-babdullakhi-sw-2(config-if)#span
msk-donskaya-babdullakhi-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-babdullakhi-sw-2(config-if)#int f0/2
msk-donskaya-babdullakhi-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-babdullakhi-sw-2(config-if)#
```

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Рис. 6: PortFast

Проверка отказоустойчивости STP

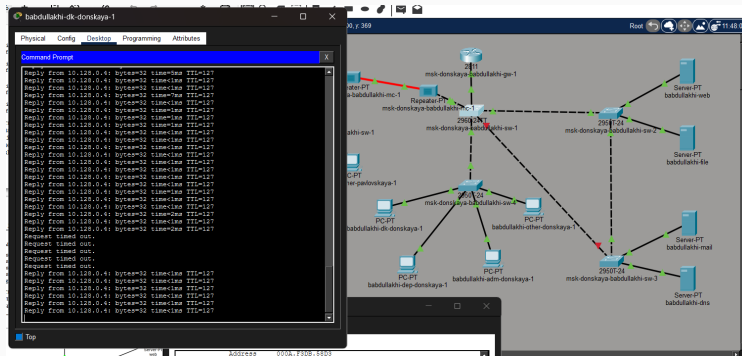


Рис. 7: STP отказ

```
msk-donskaya-babdullakhi-sw-1(config)#spa
msk-donskaya-babdullakhi-sw-1(config)#spanning-tree mode rap
msk-donskaya-babdullakhi-sw-1(config)#spanning-tree mode rapid-pvst
msk-donskaya-babdullakhi-sw-1(config)#exit
msk-donskaya-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-sw-1#
```

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Рис. 8: Rapid PVST+

Проверка Rapid PVST+

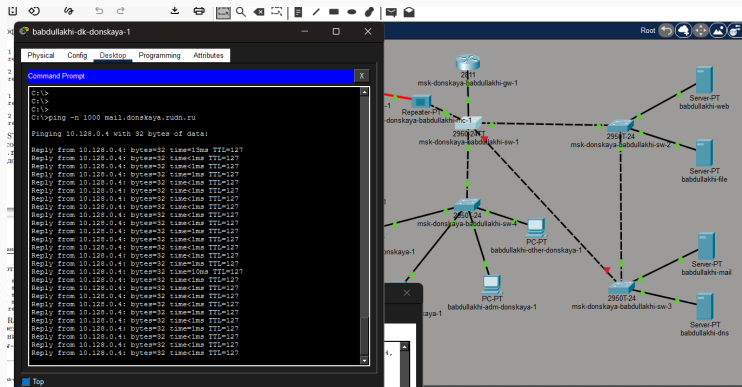


Рис. 9: Rapid PVST test

```
[OK]
msk-donskaya-babdullakhi-sw-1#
msk-donskaya-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-babdullakhi-sw-1(config)#int range f0/20 - 23
msk-donskaya-babdullakhi-sw-1(config-if-range)#channel-group 1 mode on
msk-donskaya-babdullakhi-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be
suspended (duplex of Fa0/23 is full, Fa0/20 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be
suspended (duplex of Fa0/23 is full, Fa0/21 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be
suspended (duplex of Fa0/23 is full, Fa0/22 is half)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed
state to down

msk-donskaya-babdullakhi-sw-1(config-if-range)#exit
msk-donskaya-babdullakhi-sw-1(config)#int port-channel 1
msk-donskaya-babdullakhi-sw-1(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-1(config-if)#
```

Рис. 10: EtherChannel настройка

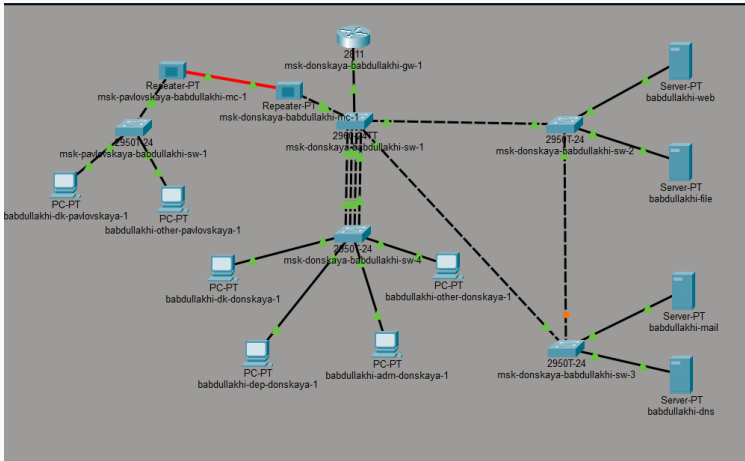


Рис. 11: EtherChannel результат

```
C:\>
C:\>ping -n 1000 mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

Control-C
^C
C:\>
```

Рис. 12: Ping результат

Выводы

Настроена отказоустойчивая сеть с использованием STP и Rapid PVST+.
Реализовано резервирование каналов и агрегирование EtherChannel.
Проверена корректная работа сети при отказах и восстановлении соединений.